

**RagForge: Evaluating Retrieval-Augmented Generation  
for Long-Document Literary Question Answering**

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## Abstract

RagForge is a retrieval-augmented generation (RAG) system built over the complete text of J.R.R. Tolkien’s *Lord of the Rings* trilogy to evaluate how RAG design choices affect grounded question answering on long narrative text. The system chunks approximately 576,000 words into 3,417 retrievable passages, embeds them with Amazon Titan v2, stores them in a Pinecone vector database, and retrieves evidence using hybrid dense–BM25 search before generating answers with Amazon Nova Pro or Claude Opus 4.6. A curated dataset of 1,000 questions (749 training, 251 held-out test), spanning factoid and multi-hop types, was used for systematic evaluation. Retrieval parameters—chunk size, retrieval depth, and hybrid weighting—were tuned using Hit@K, MRR, and nDCG. Generation parameters—temperature, output length, query expansion, and context ordering—were evaluated with both keyword accuracy and LLM-judge scoring.

The central finding is that, within this system and corpus, generator capability was a larger bottleneck than further retrieval tuning once a strong retrieval baseline had been established: on identical retrieval with a shared 89.6% hit rate, Opus 4.6 achieved 87.6% judge accuracy versus Nova Pro’s 74.5%—a 13.1-point improvement that exceeded the end-to-end gains observed from retrieval tuning in this study. Additional findings include that keyword scoring inflates accuracy by 3–5 points relative to LLM-judge evaluation, and that two independent literature-recommended techniques—query expansion and calibrated abstention—failed to improve performance on this domain-specific corpus.

## Introduction

When a language model is asked a question about a specific source text, it frequently produces responses containing hallucinated facts, conflated events, omitted context, or

unsupported claims. The problem is especially acute in long-document question answering, where relevant evidence may be scattered across hundreds of pages rather than contained in a single passage, and the model has no reliable way to distinguish what it actually learned from the source versus what it is generating from parametric memory.

Retrieval-Augmented Generation addresses this problem by retrieving relevant passages from a source text before passing them to a language model for answer generation. Rather than relying solely on parametric memory, a RAG system grounds its responses in textual evidence. Lewis et al. (2020) demonstrated that this approach outperforms purely parametric models on knowledge-intensive NLP tasks by combining a Dense Passage Retriever with a BART-based sequence-to-sequence generator, building on the dense retrieval framework of Karpukhin et al. (2020). Since then, RAG has become the dominant architecture for building question-answering systems that must remain grounded in specific documents.

Building a RAG system that works reliably on long narrative text, however, introduces challenges that standard benchmarks do not fully address. Most RAG research evaluates on short, self-contained documents such as Wikipedia articles, where each passage covers a single topic and answers tend to be concise factual statements. Thakur et al. (2021) showed through the BEIR benchmark that retrieval performance varies substantially across domains, suggesting that findings from one corpus may not transfer to another. A long narrative like *The Lord of the Rings* is fundamentally different: characters appear under multiple names, events unfold across chapters with causal relationships spanning hundreds of pages, and many questions require synthesizing information from multiple scattered passages.

Despite growing interest in RAG parameter optimization, much existing research evaluates retrieval in isolation from generation. Bhat et al. (2025) conducted a systematic

analysis of how chunk size affects retrieval effectiveness across multiple datasets, finding that chunk size significantly influences performance and that the optimal size depends on the dataset and task. Their work is an important contribution, but it relies on string-matching metrics, measures only retrieval quality without assessing end-to-end generation, and uses primarily synthetic datasets. Izacard and Grave (2021) showed through Fusion-in-Decoder that generators can leverage large numbers of retrieved passages, but their evaluation similarly focused on standard benchmarks rather than domain-specific narrative text.

This project, RagForge, addresses these gaps. It is an end-to-end RAG pipeline built over the complete *Lord of the Rings* trilogy—approximately 576,000 words across three books. The system chunks the text into retrievable passages, embeds them with Amazon Titan v2, stores them in Pinecone, and retrieves evidence using hybrid dense–BM25 search before generating answers with Amazon Nova Pro or Claude Opus 4.6.

The goal was not to build a question-answering chatbot but to systematically evaluate how specific design choices affect grounded question answering on long narrative text. To that end, I curated a dataset of 1,000 questions spanning factoid and multi-hop types and split them into 749 training and 251 held-out test questions. I ran controlled experiments across the major RAG parameters identified in the literature: chunk size, retrieval depth, hybrid search weighting, temperature, output length, query expansion, refusal thresholds, and model choice. Retrieval was evaluated with Hit@K, MRR, and nDCG; answer quality was evaluated with both keyword accuracy and LLM-judge scoring.

The central finding is that generator capability was a larger bottleneck than further retrieval tuning once a strong retrieval baseline had been established. On identical retrieval—same chunks, same hit rate, same configuration—Opus 4.6 achieved 87.6% judge

accuracy compared to Nova Pro’s 74.5%, a 13.1-point improvement that exceeded the end-to-end gains observed from retrieval tuning in this study. This suggests that for long-document literary question answering, investing in a stronger generator yields greater returns than further optimizing retrieval once a reasonable baseline is established.

## **Literature Review**

### ***Retrieval-Augmented Generation***

The concept of augmenting language model generation with retrieved evidence was formalized by Lewis et al. (2020). Their architecture combined a Dense Passage Retriever—based on the bi-encoder framework of Karpukhin et al. (2020)—with a BART-based generator trained end-to-end. The system retrieved from Wikipedia split into disjoint 100-word passages and achieved state-of-the-art results on Natural Questions, TriviaQA, and WebQuestions. Lewis et al. tested two variants: RAG-Sequence, which conditions the entire answer on a single retrieved document, and RAG-Token, which attends to different documents at each generation step. They used top-k values of 5 and 10 with greedy decoding for factoid QA and beam search for longer tasks. A key insight was that retrieval augmentation not only improved factual accuracy but also reduced hallucination by grounding generation in external evidence.

Izacard and Grave (2021) extended this work with Fusion-in-Decoder (FiD), which processes each retrieved passage independently through the encoder before fusing them in the decoder. This architecture scales to much larger passage counts—they showed accuracy improving as passage count increased from 5 to 100—establishing that generators can leverage many retrieved contexts. Their work demonstrated that the number of passages provided to the generator is a meaningful experimental variable with effects that depend on the task and model.

The RAG paradigm has since become the standard for knowledge-grounded question answering. However, the evaluation approaches used by Lewis et al. and Izacard and Grave relied primarily on Exact Match scoring, and their corpora consisted of short, self-contained Wikipedia passages—characteristics that do not represent the challenges of applying RAG to long, continuous narrative text where evidence spans chapters and questions range from simple fact lookup to multi-hop reasoning.

### ***Chunk Size as a Retrieval Variable***

Among the many tunable RAG parameters, chunk size—the length of each text segment indexed for retrieval—has emerged as one of the most studied. Bhat et al. (2025) conducted a systematic evaluation of how chunk size affects retrieval performance across multiple datasets and embedding models. They found that chunk size significantly influences retrieval effectiveness, that the optimal size depends on the dataset and task, and that smaller chunks tend to improve precision for fact-based queries while larger chunks preserve broader context for complex questions. They tested chunk sizes from 64 to 1,024 tokens and observed that the interaction between chunk size and embedding model architecture matters: decoder-based models performed better with larger chunks, while encoder-based models favored smaller ones.

However, Bhat et al.’s study has three limitations that this project addresses. First, their evaluation used string-matching metrics, which can overstate performance when a retrieved passage contains the right keywords but does not support a correct answer. Second, they evaluated retrieval in isolation without measuring whether improvements translated to better end-to-end generation. Third, they relied primarily on synthetic datasets. RagForge addresses all three by using LLM-judge evaluation alongside keyword scoring, measuring generation quality rather than retrieval alone, and evaluating on 1,000 questions written for a natural literary corpus.

### ***Hybrid Retrieval and Fusion Strategies***

A consistent finding in the RAG literature is that hybrid retrieval—combining dense semantic search with sparse keyword-based search such as BM25—outperforms either method alone. Dense retrieval excels at semantic relationships and paraphrases (Karpukhin et al., 2020), while BM25 excels at exact term matching for rare entities and proper nouns. Thakur et al. (2021) demonstrated through BEIR that BM25 remains a robust baseline across diverse domains and that retrieval performance is highly task-dependent, reinforcing the value of hybrid approaches.

Reciprocal Rank Fusion (RRF), introduced by Cormack et al. (2009), has become the standard method for combining dense and sparse results. Unlike linear score interpolation, which requires careful normalization, RRF operates on rank positions alone, making it robust across retrieval backends. The typical implementation uses a constant of  $k=60$ . In this work, hybrid retrieval merges dense and BM25 results using an RRF-style fusion scheme with  $k=60$ , and a tunable alpha parameter controls the relative contribution of dense and sparse retrieval in the final merged ranking.

### ***Context Utilization and the Lost-in-the-Middle Phenomenon***

Liu et al. (2023) demonstrated a U-shaped attention pattern in language models, where information at the beginning and end of the context receives stronger attention than information in the middle. This has direct implications for RAG: concatenating retrieved chunks by retrieval score may not be optimal if the most relevant passage lands in the middle of a long context. Izacard and Grave (2021) addressed this architecturally through independent encoding in FiD, but most production systems use simpler concatenation where ordering effects may apply.

### ***Adaptive Retrieval and Calibrated Abstention***

Recent work has explored mechanisms for RAG systems to abstain rather than risk hallucination when retrieval is insufficient. Asai et al. (2024) introduced Self-RAG, which trains the model to output special tokens governing retrieval and self-critique. Yan et al. (2024) proposed Corrective RAG (CRAG), which scores retrieved documents and triggers corrective actions when confidence is low. These approaches reflect the field’s recognition that retrieval can actively mislead generation when context is irrelevant or noisy.

### ***Evaluation Beyond Keyword Matching***

A growing body of work argues that traditional Exact Match and keyword-based F1 scores are insufficient for RAG assessment. Gao et al. (2023) formalized this in ALCE, which evaluates citation quality alongside answer correctness and showed that even factually correct answers can lack citation support. Es et al. (2024) proposed the RAGAS framework with metrics including faithfulness, answer relevancy, and context precision. LLM-as-a-judge evaluation has gained adoption as a more nuanced alternative to string matching, particularly for open-ended or multi-sentence answers common in literary QA.

### ***Gap in the Literature***

While the RAG literature provides extensive guidance on parameter tuning for standard QA benchmarks, relatively little work applies these techniques to long-form literary text. Most evaluations use Wikipedia, news articles, or technical documentation. Literary text introduces unique challenges: narrative continuity, characters with multiple names, temporal dependencies between events, and questions requiring synthesis from widely separated passages. This project contributes by applying systematic RAG parameter tuning to a literary corpus, evaluating with both retrieval metrics and end-to-end LLM-judge scoring, and testing whether standard literature

recommendations—including query expansion and calibrated abstention—transfer to this domain.

## **Methodology**

### ***Corpus Preparation***

The source corpus is the complete text of J.R.R. Tolkien’s *The Lord of the Rings* trilogy: *The Fellowship of the Ring*, *The Two Towers*, and *The Return of the King*. The combined text contains approximately 576,000 words across 62 chapters. No supplementary material—movie scripts, wiki summaries, or secondary sources—was included.

The text was split into fixed-size chunks of approximately 200 tokens each, producing 3,417 passages: 1,409 from *Fellowship*, 1,225 from *Two Towers*, and 783 from *Return of the King*. Each chunk was tagged with metadata including book name, chapter number, chapter title, and character-level positions within the source text. The corpus was also chunked at 500-token and 1,000-token granularities for controlled chunk-size experiments, following Bhat et al.’s (2025) recommendation to test multiple sizes.

### ***Embedding and Vector Storage***

Each chunk was embedded using Amazon Titan Embed Text v2 (1,024-dimensional vectors) and stored in a Pinecone vector database with cosine similarity as the distance metric, consistent with standard practice for normalized embeddings (Karpukhin et al., 2020). A BM25 index was built over the same text using parameters  $k_1=1.5$  and  $b=0.75$  for hybrid retrieval.

### ***Retrieval Pipeline***

The retrieval pipeline supports pure dense, pure BM25, and hybrid modes. Hybrid retrieval uses an RRF-style fusion (Cormack et al., 2009) with  $k=60$  to combine dense and sparse

result lists, and a tunable alpha parameter controls the relative contribution of dense and sparse retrieval in the final merged ranking. The final tuned configuration uses  $\alpha=0.8$  with  $K=20$  retrieved passages and a minimum cosine similarity threshold of 0.30.

### ***Multi-Hop Question Handling***

Questions are classified as factoid or multi-hop using regex-based heuristic patterns. Multi-hop questions are decomposed into two to three sub-questions using Llama 3.2 (3B) via AWS Bedrock. Llama 3.2 was selected for decomposition because the task is structurally simple—splitting a complex question into sub-parts—and does not require a large model, making the lightweight 3B parameter model a cost-effective choice. This decision was further supported by query expansion experiments (described in the Results section), where Llama 3.2 outperformed Nova Pro on query manipulation tasks. Each sub-question is searched independently; results are merged, deduplicated, and sorted by score. The routing itself was a meaningful design decision: early experiments showed that applying decomposition to simple factoid questions degraded accuracy, so the system routes question types through different retrieval paths.

### ***Generation Pipeline***

Two generator models were evaluated: Amazon Nova Pro and Claude Opus 4.6, both accessed via AWS Bedrock at temperature 0.0 for deterministic output. Following Lewis et al.’s (2020) finding that greedy decoding works best for factoid QA, maximum output tokens are set to 200 for factoid questions and 400 for multi-hop questions. The prompt instructs the model to answer directly with citations.

### ***Question Dataset***

The evaluation dataset consists of 1,000 curated questions split into 749 training and 251 held-out test questions (88% factoid, 12% multi-hop). Each question includes expected answer keywords for automated scoring. I established the train–test split before any parameter tuning, and the held-out set was used only for final evaluation after all configurations were locked.

The questions were generated using Claude Opus 4.6 with detailed prompts specifying question types, topic coverage across all three books, and expected answer keywords grounded in the source text. I reviewed and curated the generated questions, removing duplicates, correcting expected answers against the trilogy text, and ensuring balanced coverage across books, chapters, and question types. Expected answer keywords were designed to capture synonymous forms (e.g., “Ithilien,” “Cormallen,” “Field of Cormallen” for the same location). Because the dataset was generated by an LLM and curated by a single annotator, the results should be interpreted as a controlled but semi-automated benchmark rather than a community-standard evaluation set. No formal inter-annotator agreement was computed; however, the dataset is made available for independent validation. It should also be noted that Opus 4.6 served as both the question generation model and the LLM judge, which introduces potential circularity—the same model family’s biases could influence both the evaluation criteria and the answers being evaluated. This limitation is discussed further in the Limitations section.

### ***Evaluation Framework***

Retrieval metrics follow the recommendations of Es et al. (2024) and Thakur et al. (2021). Hit@K measures whether at least one relevant chunk appears in the top-K results. Mean Reciprocal Rank (MRR) measures how early the first relevant chunk appears. Normalized Discounted Cumulative Gain (nDCG@K) measures overall ranking quality. Relevance is approximated by checking whether a chunk contains expected answer keywords.

For generation evaluation, keyword accuracy measures whether the generated answer contains expected keywords. This metric is fast but can inflate accuracy, as Gao et al. (2023) demonstrated in their analysis of the gap between retrieval recall and generator correctness. LLM-judge accuracy uses Opus 4.6 to evaluate semantic correctness regardless of phrasing. The difference between keyword and judge accuracy—termed “inflation”—is itself an important measurement of evaluation methodology quality.

### ***Infrastructure***

The system runs on AWS: a Lambda function behind API Gateway serves the RAG pipeline, Pinecone provides vector storage, Bedrock provides model access, and a Streamlit web application provides the user interface. Evaluation notebooks run on a SageMaker instance (ml.t3.medium).

## **Results and Discussion**

All parameters were tuned on the 749-question training set. The 251-question held-out test was evaluated only after all configurations were locked.

### ***Retrieval Tuning (Training Set, $n = 749$ )***

**Chunk Size.** I compared three chunk sizes at  $\alpha=0.5$  and  $K=10$ , following Bhat et al.’s (2025) methodology.

**Table 1. Chunk Size Comparison**

| Chunk Size | Hit@10 | MRR   | nDCG@10 | Multi-hop Hit |
|------------|--------|-------|---------|---------------|
| ~200-tok   | 85.1%  | 0.581 | 0.613   | 91.3%         |
| 500-tok    | 82.2%  | 0.582 | 0.595   | 90.2%         |
| 1,000-tok  | 78.4%  | 0.519 | 0.548   | 95.7%         |

The 200-token configuration achieved the best overall hit rate and nDCG, supporting Bhat et al.’s finding that smaller chunks improve precision for fact-based queries. The 1,000-token configuration achieved the highest multi-hop hit rate (95.7%), demonstrating the tradeoff between factoid precision and multi-hop coverage. This interaction between chunk size and question type reflects a pattern Izacard and Grave (2021) also documented: different tasks benefit from different amounts of context.

**Retrieval Depth (K).** With  $\sim 200$ -tok chunks and  $\alpha=0.5$ , K was swept from 5 to 50.

**Table 2. Retrieval Depth Sweep**

| K  | Hit@K | MRR   | nDCG@K |
|----|-------|-------|--------|
| 5  | 76.0% | 0.571 | 0.583  |
| 10 | 85.1% | 0.581 | 0.613  |
| 15 | 89.3% | 0.584 | 0.638  |
| 20 | 91.1% | 0.585 | 0.641  |
| 30 | 93.3% | 0.586 | 0.642  |
| 40 | 93.7% | 0.586 | 0.641  |
| 50 | 94.5% | 0.586 | 0.640  |

nDCG peaked at K=30 and declined slightly at K=40–50. Lewis et al. (2020) observed a similar pattern, where higher top-k values introduce a tradeoff between recall and noise. K=20 was selected over K=30 because the marginal nDCG gain was negligible (0.641 vs. 0.642) while the additional 10 chunks increase context length from approximately 4,000 to 6,000 tokens, introducing more noise without meaningful retrieval improvement. At  $\sim 200$ -token chunks, K=20 keeps the total context at approximately 4,000 tokens, a range where Nova Pro demonstrated consistent generation quality. Optimal K also proved model-dependent: earlier experiments with a smaller generator showed degradation at high K, while Nova Pro benefited from K=20—a pattern Izacard and Grave (2021) also documented, where larger models handle more passages more effectively.

**Hybrid Weighting ( $\alpha$ ).** Alpha=0.8 produced the best results on the full training set with the tuned configuration:

**Table 3. Tuned Retrieval Configuration ( $\alpha = 0.8$ ,  $K = 20$ )**

| Metric        | Value |
|---------------|-------|
| Hit Rate      | 91.7% |
| MRR           | 0.594 |
| nDCG          | 0.651 |
| Factoid Hit   | 91.0% |
| Multi-hop Hit | 96.7% |

The dense-heavy hybrid outperformed both pure dense and pure BM25, validating the complementary strengths described by Karpukhin et al. (2020) and Cormack et al. (2009). That the optimal alpha (0.8) is higher than the 0.3–0.7 range typically reported suggests that Titan v2 embeddings capture LOTR-specific terminology well enough that BM25’s contribution is supplementary rather than essential.

***Generation Tuning (Training Set,  $n = 749$ )***

**Temperature.** Temperature 0.0 achieved the highest judge accuracy (80.6%) despite slightly lower keyword accuracy than 0.1. Lewis et al. (2020) used greedy decoding for factoid QA for similar reasons.

**Table 4. Temperature Comparison**

| Temperature | Keyword | Judge |
|-------------|---------|-------|
| 0.0         | 80.2%   | 80.6% |
| 0.1         | 80.6%   | 78.9% |

**Maximum Output Tokens.** The 200/400 configuration (factoid/multi-hop) achieved the best judge accuracy (80.5%). A 400/600 configuration had the highest keyword accuracy

(81.6%) but worst judge accuracy (78.5%), revealing a 3.1-point inflation gap: longer answers contain more keywords without being more correct.

**Table 5. MaxTokens Comparison**

| Factoid / Multi-hop | Keyword | Judge Overall | Judge Factoid | Judge Multi-hop |
|---------------------|---------|---------------|---------------|-----------------|
| 200 / 400           | 81.0%   | 80.5%         | 80.1%         | 83.7%           |
| 150 / 500           | 80.2%   | 80.2%         | 79.8%         | 83.7%           |
| 200 / 500           | 80.0%   | 79.7%         | 79.0%         | 84.8%           |
| 300 / 500           | 80.0%   | 79.6%         | 78.8%         | 84.8%           |
| 400 / 600           | 81.6%   | 78.5%         | 78.1%         | 81.5%           |

**Context Ordering.** Score-ranked versus chapter-position ordering showed a +0.1% difference on 749 questions—negligible. Despite Liu et al.’s (2023) Lost-in-the-Middle findings, passage ordering did not meaningfully affect generation quality in these experiments.

**Query Expansion.** Llama 3.2 expansion produced a +2% hit rate improvement but flat end-to-end accuracy. Nova Pro expansion reduced accuracy by 2.4%. The expanded queries introduced semantic drift—additional terms that broadened retrieval without improving relevance. The literature cautions that query expansion can improve recall but worsen precision when expansions drift from the original intent, and that is what happened here.

**Refusal Threshold.** Score-based refusal failed because top retrieval scores were too tightly clustered to serve as a useful confidence signal, unlike the dataset-specific thresholds reported by Yan et al. (2024) in CRAG. Calibrated abstention—by any method tested—does not currently work for this system.

### ***Held-Out Test Results ( $n = 251$ )***

After locking all configurations on the training set, I evaluated both models on the 251-question held-out test—the first and only time these questions were used.

**Table 6. Held-Out Test: Nova Pro vs. Opus 4.6**

| Metric                 | Nova Pro | Opus 4.6 | $\Delta$ |
|------------------------|----------|----------|----------|
| Keyword Overall        | 79.3%    | 90.4%    | +11.1%   |
| Keyword Factoid        | 78.2%    | 90.5%    | +12.3%   |
| Keyword Multi-hop      | 87.1%    | 90.3%    | +3.2%    |
| Judge Overall          | 74.5%    | 87.6%    | +13.1%   |
| Judge Factoid          | 74.1%    | 86.4%    | +12.3%   |
| Judge Multi-hop        | 77.4%    | 96.8%    | +19.4%   |
| Hit Rate               | 89.6%    | 89.6%    | same     |
| Inflation (kw – judge) | +4.8%    | +2.8%    | —        |

This is the central result. On identical retrieval—same chunks, same configuration, same 89.6% hit rate—Opus outperformed Nova Pro by 13.1 points on judge accuracy. The improvement was particularly dramatic for multi-hop questions (96.8% vs. 77.4%), suggesting that stronger models excel at synthesizing evidence across multiple passages. Izacard and Grave (2021) documented the same dynamic: model capability, not retrieval volume, determines how effectively retrieved evidence is utilized.

**Error Taxonomy.** A manual classification of errors on the held-out test reveals the distribution of failure modes. The categories are mutually exclusive; I classified each error by reviewing the retrieved chunks, generated answer, and expected answer.

**Table 7. Error Taxonomy (Held-Out Test)**

| Error Category   | Count | % of Test | Description                                    |
|------------------|-------|-----------|------------------------------------------------|
| Noisy Context    | 23    | 9.2%      | Right chunks retrieved, wrong answer generated |
| Judge Mismatch   | 24    | 9.6%      | Keyword says correct, judge says incorrect     |
| Keyword Mismatch | 9     | 3.6%      | Answer semantically correct, keyword missed    |
| Retrieval Miss   | 10    | 4.0%      | Relevant passage not in retrieved set          |

Generation failures (noisy context, 9.2%) outnumber retrieval failures (4.0%) by more than 2:1. This finding directly motivated the model comparison: Opus’s superior performance

supports the interpretation that a stronger generator reduces noisy-context errors. The judge-mismatch category (9.6%) quantifies keyword inflation—cases where answers contained expected keywords but were not actually correct. Gao et al. (2023) argued that string-matching evaluation is insufficient for exactly this reason.

**Keyword Inflation.** A practitioner evaluating Nova Pro with keyword scoring alone would conclude the system answers roughly 4 out of 5 questions correctly. The LLM judge reveals the actual rate is closer to 3 out of 4. This 5-point gap is large enough to change deployment decisions and raises questions about RAG evaluation practices in the broader literature, where keyword or exact-match scoring is often the only reported metric.

**Training vs. Test Comparison.** The table below compares Nova Pro performance across training and test sets, followed by the Opus held-out results. Opus was evaluated only on the held-out test as a controlled comparison using the same locked retrieval configuration.

**Table 8. Training vs. Test Generalization (Nova Pro)**

| Metric             | Training (n=749) | Test (n=251) | $\Delta$ |
|--------------------|------------------|--------------|----------|
| Retrieval Hit Rate | 91.7%            | 89.6%        | −2.1%    |
| Keyword Accuracy   | 81.0%            | 79.3%        | −1.7%    |
| Judge Accuracy     | 80.5%            | 74.5%        | −6.0%    |

The modest keyword accuracy drop (−1.7%) and hit rate drop (−2.1%) indicate that retrieval and generation configurations generalize well to unseen questions. The larger judge accuracy drop (−6.0%) warrants closer examination. Keyword accuracy, which rewards any mention of expected terms, shows consistent performance across splits—suggesting retrieval quality and basic generation capability transfer. The judge, however, evaluates synthesis quality: whether the answer correctly integrates evidence, avoids conflation, and addresses the question precisely. Nova Pro’s generation ceiling becomes more visible on held-out questions where the

model encounters unfamiliar evidence combinations. This interpretation is supported by the Opus comparison: Opus shows only a +2.8% inflation gap on the same test set, indicating that the judge drop is a generator limitation rather than a retrieval or dataset artifact.

**Table 9. Held-Out Test: Opus 4.6**

| Metric          | Value |
|-----------------|-------|
| Keyword Overall | 90.4% |
| Judge Overall   | 87.6% |
| Judge Multi-hop | 96.8% |
| Inflation       | +2.8% |

Opus’s lower inflation (+2.8% vs. +4.8%) indicates that its answers are not only more accurate but more honestly represented by keyword scoring—a stronger model produces answers that are correct for the right reasons rather than incidentally containing expected keywords.

### ***Limitations***

Several limitations should be acknowledged. First, the relevance judgments used for retrieval metrics are based on keyword presence, which may overestimate relevance if a chunk contains a keyword but not the actual answer context. Second, the LLM judge (Opus 4.6) was not validated against human annotators; future work should include a human evaluation sample to confirm the judge’s reliability. Third, Opus 4.6 served as the question generator, the LLM judge, and one of the two evaluated generators, introducing potential circularity: the same model family’s biases could influence question framing, evaluation criteria, and answer quality simultaneously. The keyword-independent metrics (hit rate, MRR, nDCG) and the keyword accuracy scores—which do not involve the judge—provide partial mitigation, as they show the same directional patterns. Fourth, Opus was evaluated only on the held-out test set rather than used for full training sweeps, due to Bedrock rate limits and the substantially higher inference

cost of Opus relative to Nova Pro; a full Opus training sweep would have required significantly more time and budget than was available. Fifth, the study uses only two closed-source generator models; the generator capability finding should be tested on open-weight models to assess generalizability. Sixth, the question dataset was generated by an LLM and curated by a single reviewer without formal inter-annotator agreement, which may introduce biases in question framing or expected answers. Finally, the work is confined to a single literary work; whether the findings transfer to other long-form texts remains an open question.

## **Conclusions and Future Work**

### ***Conclusions***

The most significant finding is that, within this system and corpus, generator capability was a larger bottleneck than further retrieval tuning once a strong retrieval baseline had been established. On the held-out test, Opus 4.6 achieved 87.6% judge accuracy compared to Nova Pro’s 74.5%—a 13.1-point improvement on identical retrieval. For multi-hop questions, Opus reached 96.8% versus Nova Pro’s 77.4%. This suggests that once a reasonable retrieval baseline is established, investing in a stronger generator yields greater returns than further optimizing retrieval parameters. Izacard and Grave (2021) reached a similar conclusion from the opposite direction, showing that model capability is the primary factor in how effectively retrieved evidence gets used.

Evaluation methodology matters as much as system design. Keyword scoring consistently overstated performance by 3–5 points compared to LLM-judge evaluation—a gap that Gao et al. (2023) and Es et al. (2024) warned about in their own evaluations. The use of proper retrieval metrics—Hit@K, MRR, and nDCG as recommended by Thakur et al.

(2021)—alongside judge scoring provided a more complete picture of system behavior than any single metric.

Two independent techniques recommended by the literature—query expansion and score-based refusal thresholds—failed to improve performance on this domain. Query expansion introduced semantic drift without improving accuracy. Score-based refusal thresholds (Yan et al., 2024) failed because retrieval scores were too tightly clustered. These results suggest that standard RAG optimization techniques do not automatically transfer to domain-specific literary question answering, and serve as a cautionary finding for practitioners applying literature recommendations without domain validation.

Retrieval design still matters within a given generator. The tuned configuration (~200-tok, K=20, hybrid  $\alpha=0.8$ ) achieved a 91.7% hit rate—substantially better than untuned baselines. Hybrid retrieval validated the complementary strengths of dense and sparse search described by Karpukhin et al. (2020) and Cormack et al. (2009), and the multi-hop routing decision—sending factoid and multi-hop questions through different retrieval paths—prevented decomposition from degrading simple questions.

### ***Future Work***

Applying the pipeline to another long narrative such as Homer’s *Odyssey* would test whether these findings generalize beyond Tolkien’s prose. Structure-aware chunking that respects paragraph and scene boundaries could address fragmentation at chunk borders. Dual-granularity retrieval—merging 200-token and 1,000-token indexes—could combine factoid precision with multi-hop context coverage. Given that this project’s central finding identifies generation as the primary bottleneck, fine-tuning a smaller open-weight generator (e.g., Llama 3.1 8B via LoRA) on gold-standard answers produced by Opus could be a particularly

high-leverage direction—potentially closing the 13-point gap between Nova Pro and Opus at a fraction of the inference cost. On the retrieval side, fine-tuning the embedding model on LOTR-specific question–answer pairs could improve domain retrieval, as Thakur et al. (2021) showed that domain adaptation significantly affects retrieval quality. Finally, chunk overlap tuning at 10–20% could reduce the boundary-loss failures observed in the error taxonomy.

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